

What is Data ?

Data is collection of raw material in unorganized format which refers to an object and it can be anything. Data is a description of things, events or activities in transactions or any system. Anything which is recorded which defines a particular event or a particular place or particular activity or a transaction are called data.

Example:-

- You just open a notepad and write something that is unorganized data.

Based on the format in which you are receiving we can categorize it into either organized format or unorganized format.

What is information?

The data which we have received, when organized in a meaningful way and has value that is called as information.

Information is the organized data with which we can make out meaning and has value to it.

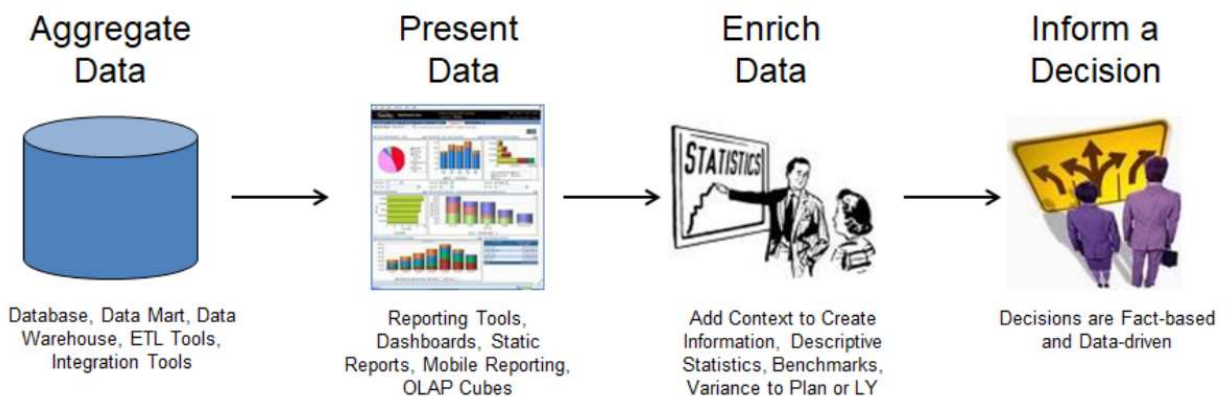
Now when you have the data and information we will get the valuable information that conveys understanding or learning applicable to a problem or activity .

Once you have the data which is organized you can process it in a specific manner using ETL tools to get knowledge insights of the data.

Now with the data in hand it is serving the purpose but you have to see if it is meeting the criteria with which the organization is getting benefited out of it.

What is Business Intelligence?

Business intelligence, or BI, is an umbrella term that refers to a variety of software applications used to analyze an organization's raw data. BI as a discipline is made up of several related activities, including data mining, online analytical processing, querying and reporting.



What is OLTP?

OLTP (online transaction processing) is a class of software programs capable of supporting transaction-oriented applications over the network.

Typically, OLTP systems are used for order entry, financial transactions, customer relationship management (CRM) and retail sales. Such systems have a large number of users who conduct short transactions. Database queries are usually simple, require sub-second response times and return relatively few records.

Example:-

- Customer-facing personnel, clerks, online shoppers



What is OLAP?

OLAP(Online Analytical Processing) consists of a type of software tools that are used for data analysis for business decisions.

Typically, OLAP provides an environment to get insights from the database retrieved from multiple database systems at one time.

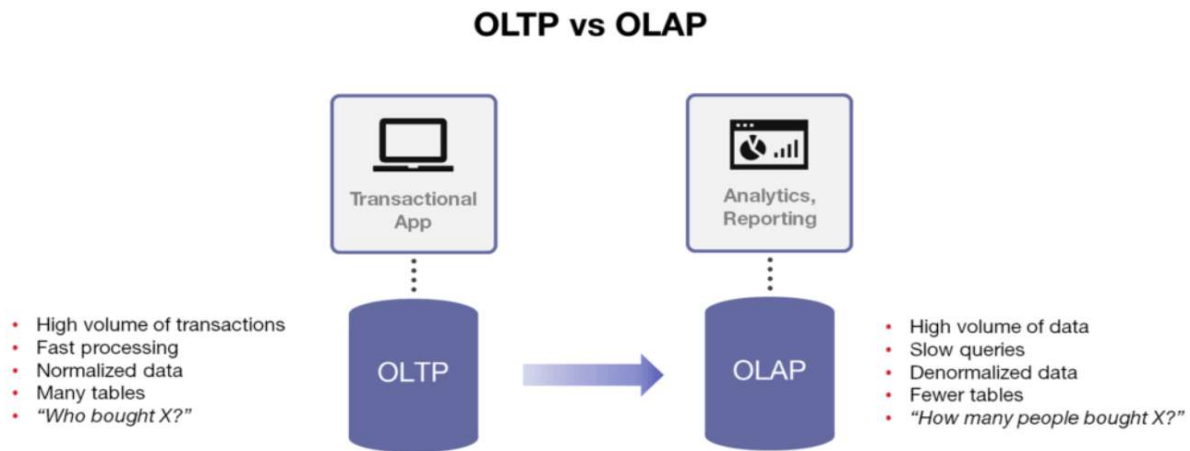
Examples:-

- Any type of Data warehouse system is an OLAP system.

Uses of OLAP are as follows:

- YouTube Music analyzed songs by users to come up with the personalized homepage of their songs and playlist.
- Netflix movie recommendation system.

OLTP VS OLAP



Properties	OLTP	OLAP (Data Warehouse)
Operations	DML	Primarily read only
Nature of Data	Operational	Aggregated, Snapshots over time
Data Organization	Applications wise	Subject, time wise
Size	Small to large	Large to very large
Data Source	Operational, Internal	Operational, Internal, External
Activities	Processes	Analysis

History of Data Warehouse

The term "Data Warehouse" was first coined by Bill Inmon in 1990. According to Inmon, a data warehouse is a subject oriented, integrated, time-variant, and non-volatile collection of data. This data helps analysts to make informed decisions in an organization.

An operational database undergoes frequent changes on a daily basis on account of the transactions that take place.

Suppose a business executive wants to analyze previous feedback on any data such as a product, a supplier, or any consumer data, then the executive will have no data available to analyze because the previous data has been updated due to transactions.

A data warehouse provides us generalized and consolidated data in a multidimensional view. Along with a generalized and consolidated view of data, a data warehouse also provides us

Online Analytical Processing OLAP tools. These tools help us in interactive and effective analysis of data in a multidimensional space. This analysis results in data generalization and data mining.

Data mining functions such as association, clustering, classification, prediction can be integrated with OLAP operations to enhance the interactive mining of knowledge at multiple levels of abstraction. That's why data warehouses have now become an important platform for data analysis and online analytical processing.

Key terms explanation

Data clustering

It is the task of grouping a set of objects in such a way that objects in the same group (called a cluster) are more similar (in some sense) to each other than to those in other groups (clusters). Clustering is grouping data points in creating partitions based on similarity.

Example:-

- Hotstar
- Streaming services often use clustering analysis to identify viewers who have similar behavior. A streaming service may collect the following data about individuals:
 - Minutes watched per day
 - Total viewing sessions per week
 - Number of unique shows viewed per month

Using these metrics, a streaming service can perform cluster analysis to identify high usage and low usage users so that they can know who they should spend most of their advertising dollars on.

Data classification

Data classification is the process of analyzing structured or unstructured data and organizing it into categories based on file type, contents, and other metadata.

Data classification helps organizations answer important questions about their data that inform how they mitigate risk and manage data governance policies. It can tell you where you are storing your most important data or what kinds of sensitive data your users create most often.

Need of data classification

Imagine you're the CEO of a 10,000-person organization where users create millions of files and emails each day. Some of that information is highly sensitive—if leaked or stolen, you're facing a headline-making breach and seven-figure penalties. Most of the data created each day, however, could be published on the front page of the Times without incident.

It can be virtually impossible to prioritize risk mitigation or comply with privacy laws when you don't know which information calls for military-grade protection. That's where data classification comes in.

Data Mining

Data mining is the process of sorting through large data sets to identify patterns and relationships that can help solve business problems through data analysis. Data mining techniques and tools enable enterprises to predict future trends and make more-informed business decisions.

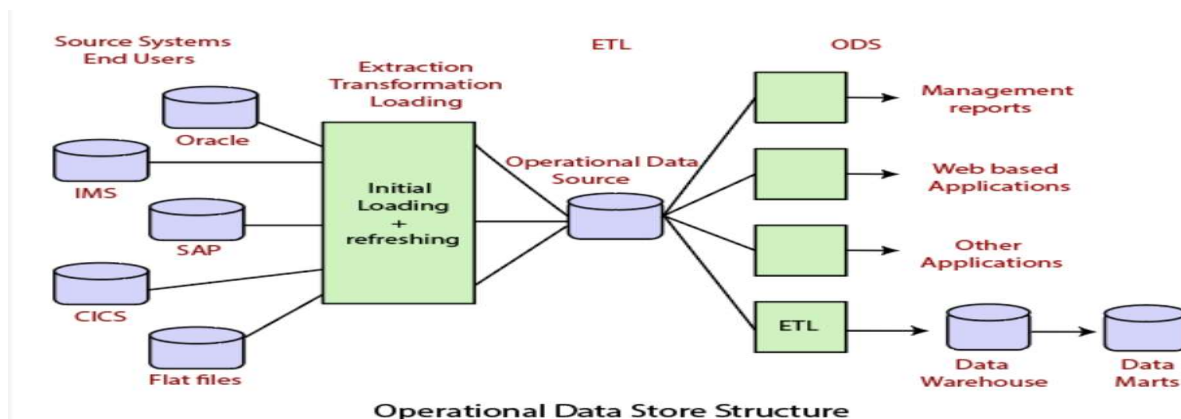
Prediction

A prediction is what someone thinks will happen. So a prediction is a statement about the future. It's a guess, sometimes based on facts or evidence, but not always.

Understanding a Data Warehouse

- A data warehouse is a database, which is kept separate from the organization's operational database.
- Data warehouse combines data from multiple variable (not static) sources into one database which can be easily manipulated. This can be used for analysis, transformation and reporting.
- There is no frequent updating done in a data warehouse.
- It possesses consolidated historical data, which helps the organization to analyze its business.
- A data warehouse helps executives to organize, understand, and use their data to make strategic decisions.
- Data warehouse systems help in the integration of diversity of application systems.
- A data warehouse system helps in consolidated historical data analysis.

ODS



In a huge organization there are different content generation tools like Oracle, SAP, Flat files, CICS systems and there is a possibility that one of the system is configured on oracle other on SQL Server ,which means you have all kind of data now this data is very business critical and you actually need to analyze this particular data to make logical decisions so your ODS or operational data store is a database where you actually integrate data from all these source systems after cleaning it based on certain business rules and you persist this data for limited time period.

Note:- ODS is not a replacement for data warehouse .It actually feeds to data warehouse and data marts. ODS actually can act as a source for data warehouses and data marts.

ODS Properties

Integrated

ODS contains an integrated version of source system so it's an integrated database it has data from different source system

Persisted (To an Extent)

ODS will not contain historic data and there is also a possibility that in certain cases it will drop data after a particular duration .

Business Rules (ETL)

All the source system data is actually cleaned using an ETL process so the data we have in our ODS system complies to certain basic rules so the data in our ODS is not raw rather it's been aligned based on certain business rules .

ODS primarily contains the current values; it actually contains the active snapshot of the system.

Example:- Being an employee in an organization I have worked on multiple projects and with different team.While in case of ODS it will only contain my current team and project information.

Partial scope

While my enterprise data warehouse is a single version of truth and it contains the entire data for the organization from all the departments that may or may not be the case with ODS. Usually the projects that I have worked on ODS only contained a subset of the data for that particular organization but there are systems with extended scope. It actually only contains the active current snapshot. As we saw in the diagram ODS can be used to push data to the mart and warehouse ,usually we insert data in our data warehouse as a part of the batch process while for ODS as per our business requirement we can actually load it multiple times

during the day or we can update the data constantly.

Data Warehouse Features

The key features of a data warehouse are discussed below:

1. **Subject Oriented** - A data warehouse is subject oriented because it provides information around the subject rather than the organization's ongoing operations. These subjects can be product, customers, suppliers, sales, revenue, etc. A data warehouse does not focus on the ongoing operations, rather it focuses on modeling and analysis of data for decision making.
2. **Integrated** - A data warehouse integrates data from multiple data sources. For example, source A and source B may have different ways of identifying a product, but in a data warehouse, there will be only a single way of identifying a product.
3. **Time Variant** - Historical data is kept in a data warehouse. For example, one can retrieve data from 3 months, 6 months, 12 months, or even older data from a data warehouse. This contrasts with a transactions system, where often only the most recent data is kept. For example, a transaction system may hold the most recent address of a customer, where a data warehouse can hold all addresses associated with a customer.
4. **Non-volatile** - Once data is in the data warehouse, it will not change. So, historical data in a data warehouse should never be altered.

What are the advantages of a data warehouse?

1. **Structured data** - Irrespective of what structure source system has, we are making our target system structured and exactly what business is looking for. As business is one of the key decision makers or requirement drivers who will give you how they want to look into the data. We as data warehouse professionals suggest them a better option.
2. **Better response time** - As the data is structured in a data warehouse you will have a better response time, which depends on volume, partition and indexing.
3. **Available historical data** - The available business data for your analysis. Historical data will be available as per the retention policy applied.
4. **Options for Ad-Hoc queries** - SQL queries can be used to analyze data as per business needs. And the majority of business tools which are available in the market do data visualization process with the GUI options. Where we don't even need to write queries.

Data Warehouse Applications

As discussed before, a data warehouse helps business executives to organize, analyze, and use their data for decision making.

Data warehouses are widely used in the following fields:

1. Financial services
2. Banking services
3. Consumer goods
4. Retail sectors
5. Controlled manufacturing

Types of Data Warehouse

Information processing, analytical processing, and data mining are the three types of data warehouse applications that are discussed below:

1. **Information Processing** - A data warehouse allows processing of the data stored in it. The data can be processed by means of querying, basic statistical analysis, reporting using crosstabs, tables, charts, or graphs.
2. **Analytical Processing** - A data warehouse supports analytical processing of the information stored in it. The data can be analyzed by means of basic OLAP operations, including slice and-dice, drill down, drill up, and pivoting.
3. **Data Mining** - Data mining supports knowledge discovery by finding hidden patterns and associations, constructing analytical models, performing classification and prediction. These mining results can be presented using the visualization tools.

ODS VS DWH

Operational Data Stores	Data warehouse
ODS stands for operational reporting and supports current or near real-time reporting requirements.	A data warehouse is intended for historical and trend analysis, usually reporting on a large volume of data.
An ODS consists of only a short window of data.	A data warehouse includes the entire history of data.
It is used for detailed decision making and operational reporting.	It is used for long term decision making and management reporting.
It is used at the operational level.	It is used at the managerial level.
It serves as conduct for data between operational and analytics systems.	It serves as a repository for cleansed and consolidated data sets.
It is updated often as the transactions system generates new data.	It is usually updated in batch processing mode on a set schedule.

Data Mart

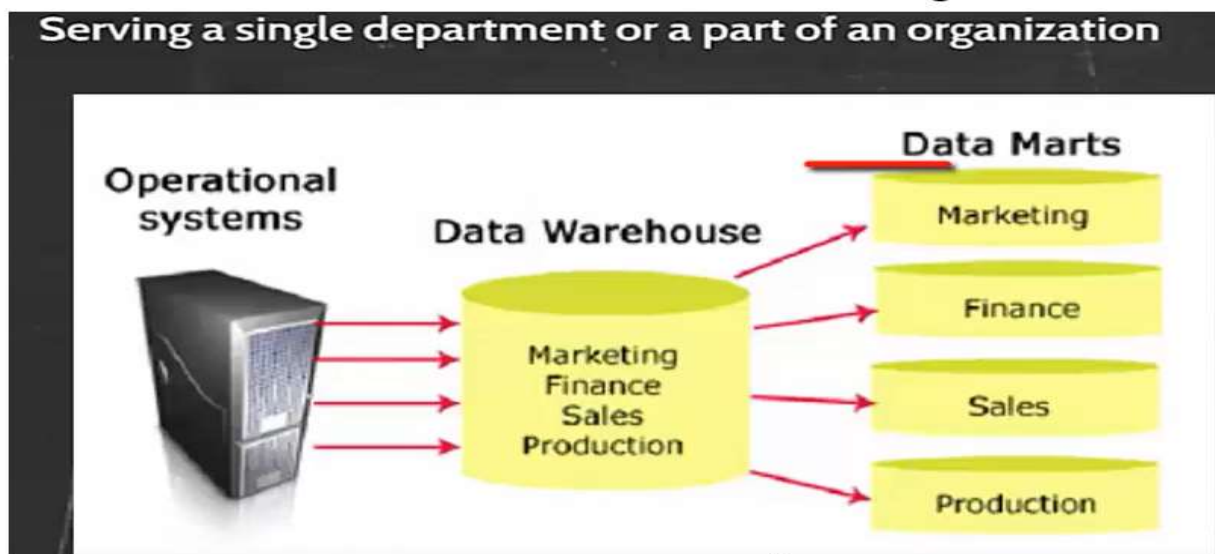
A data mart is defined as an implementation of a data warehouse with small and more tightly restricted scope of a data and data warehouse functions which are serving a single department or a part of an organization, that means it is a subset of a complete data warehouse.

Thus, the primary purpose of a data mart is to isolate—or partition—a smaller set of data from a whole to provide easier data access for the end consumers.

Example:-

Amazon being an organization has its multiple departments, where each department has their own relational database to manage their data as individual databases.

As a whole we need a data warehouse to manage all of this.



The fundamental difference between a data warehouse and data marts is scope as what is covered within the data mart and what is covered within the data warehouse.

A data warehouse is defined for covering all possible corporate support processes.

It is a combination of all departments, whereas a data mart covers a specific business analysis or reporting a specific problem within an area or within a department.

Advantages of a data mart

1. Close match with data requirements
2. Smaller scope
3. Visualization
4. Linked with other BI tools
5. Simpler and faster implementation

We are focussing on a small chunk of problem for a huge organization, so whenever we have the details for which we are looking for or details for the smaller set the output will always be better because scope is reduced to some certain level of analysis, certain level of processing which we have to do. Since data mart has a limited scope than the data warehouse they usually can be established quite faster and can therefore generate quicker return on investment.

If you plan on implementing a data warehouse, there are lots of things to be considered.

However for a data mart you are focussing on a small part or a small chunk of the whole data warehouse which means obviously it will be quite faster and the results which you can see can be implemented for other data marts as well.

Data marts can be used proficiently for proving end users that data for the organizations when it is coupled with good tools like the reporting tools or business objects or any of data visualization tools that can be very beneficial. It will give you insights of what is available and what is lacking behind on what they should focus on and how to improve the overall process or overall system or overalls sales for that matter to increase the profits.

As mentioned in the visualization part as we are linking into the visualization tool, you can integrate well with any specific functionality supported by any of these tools to improve the overall aspect of the business questions like what if ?



Data Mart vs Data Warehouse

Data marts and data warehouses are both highly structured repositories where data is stored and managed until it is needed. However, they differ in the scope of data stored: data warehouses are built to serve as the central store of data for the entire business, whereas a data mart fulfills the request of a specific division or business function. Because a data warehouse contains data for the entire company, it is best practice to have strict control who can access it. Additionally, querying the data you need in a data warehouse is an incredibly difficult task for the business. Thus, the primary purpose of a data mart is to isolate—or partition—a smaller set of data from a whole to provide easier data access for the end consumers.

Data Lake vs Data Warehouse

A data lake allows organizations to store large amounts of structured and unstructured data (for example, from social media or clickstream data), and to immediately make it available for real-time analytics, data science, and machine learning use cases. With a data lake, data is ingested in its original form, without alteration.

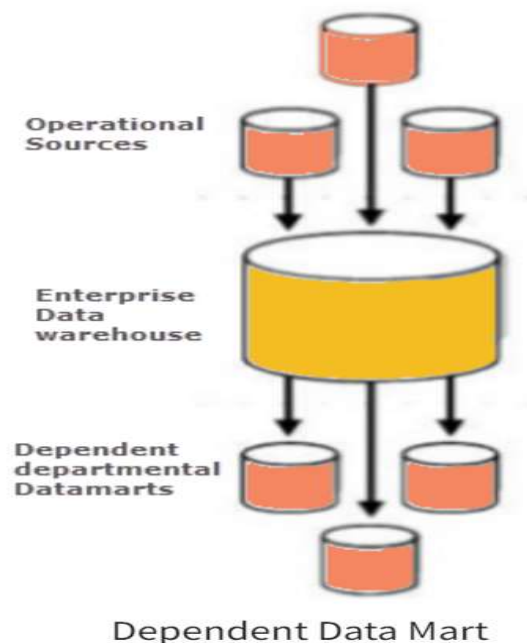
The key difference between a data lake and a data warehouse is that data lakes store vast amounts of raw data, without a predefined structure. Organizations do not need to know in advance how the data will be used.

Types of Data Mart

There are three main types of data mart:-

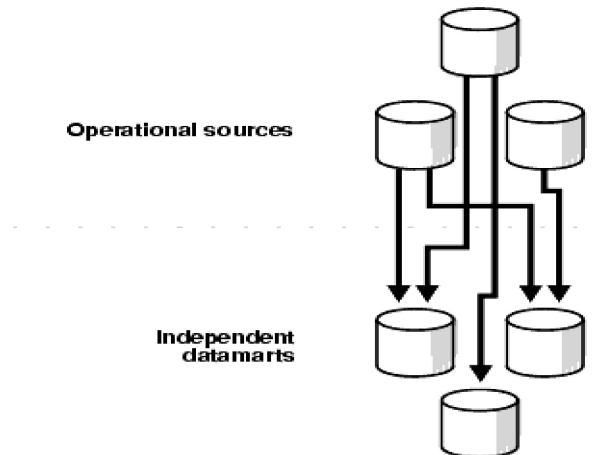
1. **Dependent:** Dependent data marts are created by drawing data directly from operational, external or both sources.
2. **Independent:** Independent data mart is created without the use of a central data warehouse.
3. **Hybrid:** This type of data marts can take data from data warehouses or operational systems.

Dependent Data Mart



Independent Data Mart

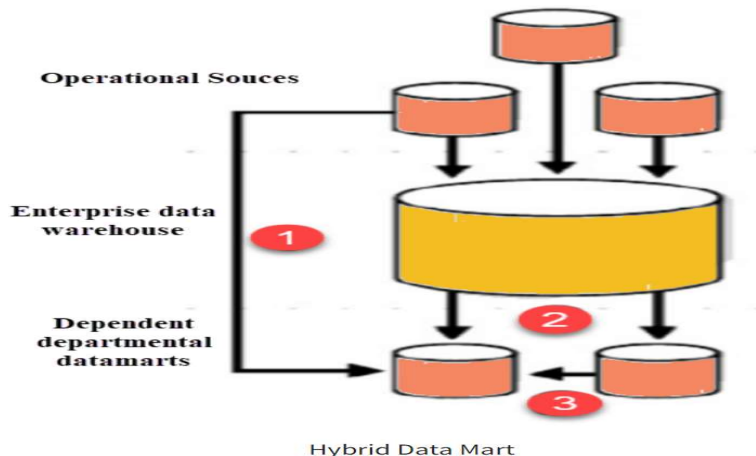
An independent data mart is created without the use of a central Data warehouse. This kind of Data Mart is an ideal option for smaller groups within an organization. An independent data mart has neither a relationship with the enterprise data warehouse nor with any other data mart. In Independent data mart, the data is input separately, and its analyses are also performed autonomously.



Hybrid Data Mart

A hybrid data mart combines input from sources apart from Data warehouse. This could be helpful when you want ad-hoc integration, like after a new group or product is added to the organization.

It is the best data mart example suited for multiple database environments and fast implementation turnaround for any organization. It also requires least data cleansing effort. Hybrid Data mart also supports large storage structures, and it is best suited for flexibility for smaller data-centric applications.



Advantages and Disadvantages of a Data Mart

Advantages

1. Data marts contain a subset of organization-wide data. This Data is valuable to a specific group of people in an organization.
2. It is a cost-effective alternative to a data warehouse, which can take high costs to build.
3. Data Mart allows faster access of Data.
4. Data Mart is easy to use as it is specifically designed for the needs of its users. Thus a data mart can accelerate business processes.
5. Data Marts need less implementation time compared to Data Warehouse systems. It is faster to implement Data Mart as you only need to concentrate on the only subset of the data.
6. It contains historical data which enables the analyst to determine data trends.

Disadvantages

1. Many times enterprises create too many disparate and unrelated data marts without much benefit. It can become a big hurdle to maintain.
2. Data Mart cannot provide company-wide data analysis as their data set is limited.

Data Warehouse VS Data Mart

Parameter	Data Warehouse	Data Mart
Definition	A Data Warehouse is a large repository of data collected from different organizations or departments within a corporation.	A data mart is an only subtype of a Data Warehouse. It is designed to meet the needs of a certain user group.
Usage	It helps to make a strategic decision.	It helps to make tactical decisions for the business.
Objective	The main objective of Data Warehouse is to provide an integrated environment and coherent picture of the business at a point in time.	A data mart mostly used in a business division at the department level.
Designing	The designing process of Data Warehouse is quite difficult.	The designing process of Data Mart is easy.
	May or may not be used in a dimensional model. However, It can feed dimensional models.	It is built focused on a dimensional model using a star schema.

Data Handling	Data warehousing includes large area of the corporation which is why it takes a long time to process it.	Data marts are easy to use, design and implement as it can only handle small amounts of data.
Focus	Data warehousing is broadly focused in all the departments. It is possible that it can even represent the entire company.	Data Mart is subject-oriented, and it is used at a department level.
Data type	The data stored inside the Data Warehouse are always detailed when compared with data mart.	Data Marts are built for particular user groups. Therefore, data is short and limited.
Scope	Data warehousing is more helpful as it can bring information from any department.	Data mart contains data of a specific department of a company. There are maybe separate data marts for sales, finance, marketing, etc. Has limited usage
Size	The size of the Data Warehouse may range from 100 GB to 1 TB+.	The Size of Data Mart is less than 100 GB.
Implementation time	The implementation process of Data Warehouse can be extended from months to years.	The implementation process of Data Mart is restricted to a few months.